



A Level • Cambridge (CIE) • Chemistry

38 mins 20 questions

Multiple Choice Practice

Topical Practice MCQ

6. Electrochemistry

Carefully selected past paper questions by topic.

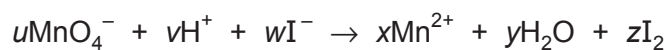
13 W moles of HNO_2 undergoes a disproportionation reaction to produce U moles of HNO_3 and V moles of NO.

- No other nitrogen containing product is produced.
- Nitrogen is the only element oxidised or reduced.

What are the values of W, U and V?

	W	U	V
A	2	1	1
B	3	1	2
C	5	3	2
D	5	1	4

- 2 The ionic equation shows iodide ions reacting with manganate(VII) ions in acidic solution.

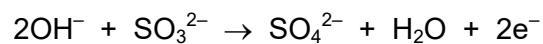
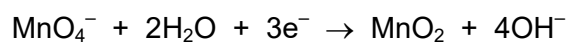


The letters u , v , w , x , y and z all represent whole numbers. Two or more of u , v , w , x , y and z are the same as each other.

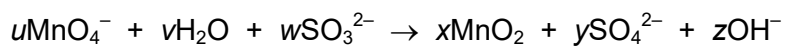
What is the lowest possible value of v ?

- A** 2 **B** 8 **C** 10 **D** 16

10 Two half-equations are shown.



The equation for the reaction between manganate(VII) ions and sulfite ions is shown.



Which statements are correct?

- 1 $u = x = z$
- 2 Manganese is reduced to oxidation state +4.
- 3 Sulfur is oxidised from oxidation state +4 to +6.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

11 Ammonium ions are converted into nitrate ions by bacteria.

What is the change in the oxidation number of nitrogen?

A -6

B +6

C +8

D +9

Question 5

9701_w22_qp_12 • Q5

- 5 Two moles of VO_2^+ ions react with one mole of zinc atoms in the presence of dilute acid. The products include Zn^{2+} ions and an ion, Y. Ion Y contains vanadium. Only zinc and vanadium change oxidation state in the reaction.

What is ion Y?

- A VO_3^- B VO^+ C VO^{2+} D VO_2^{2+}

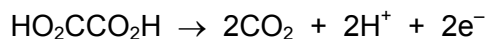
- 9** Copper dissolves in dilute nitric acid producing a blue solution of $\text{Cu}(\text{NO}_3)_2$, water and nitrogen(II) oxide as the only products.

How many moles of acid react with three moles of copper in the balanced equation?

- A** 2 **B** 4 **C** 6 **D** 8

- 9 Ethanedioic acid, $\text{HO}_2\text{CCO}_2\text{H}$, can be oxidised by KMnO_4 in dilute sulfuric acid. The products of this reaction are carbon dioxide, water, potassium sulfate and manganese(II) sulfate.

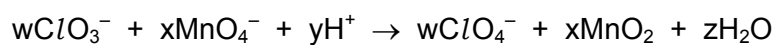
In this reaction each ethanedioic acid molecule loses two electrons as it is oxidised. A half-equation for this process is shown.



How many water molecules are produced when five ethanedioic acid molecules are oxidised by KMnO_4 in dilute sulfuric acid?

- A** 5 **B** 8 **C** 10 **D** 16

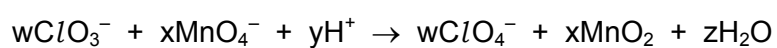
- 9 In the chemical equation, w, x, y and z are all whole numbers.



When the equation is balanced, what are w, x and y?

	w	x	y
A	1	1	2
B	2	2	2
C	2	3	8
D	3	2	2

- 9 In the chemical equation, w, x, y and z are all whole numbers.



When the equation is balanced, what are w, x and y?

	w	x	y
A	1	1	2
B	2	2	2
C	2	3	8
D	3	2	2

- 9 Oxidation numbers should be used to answer this question.

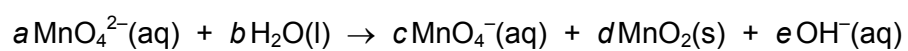
A redox reaction takes place between hydroxylammonium ions, $[\text{NH}_3\text{OH}]^+$, and acidified iron(III) ions, Fe^{3+} . The products are iron(II) ions, Fe^{2+} , H^+ ions, water and a compound of nitrogen.

The mole ratio of reacting hydroxylammonium ions to reacting iron(III) ions is 1 : 2.

Which nitrogen-containing compound could be formed in the reaction?

- A** NH_3 **B** N_2O **C** NO **D** NO_2

- 9 When K_2MnO_4 is dissolved in water, the following reaction occurs.



What are the values of a and c in the balanced chemical equation?

	a	c
A	2	1
B	3	2
C	4	3
D	5	4

22 Which pair of reagents will take part in a redox reaction?

- A** $\text{CH}_3\text{CHCH}_2 + \text{Br}_2$
- B** $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH} + \text{concentrated H}_3\text{PO}_4$
- C** $\text{CH}_3\text{COCH}_3 + \text{HCN}$
- D** $\text{HCO}_2\text{C}_2\text{H}_5 + \text{dilute H}_2\text{SO}_4$

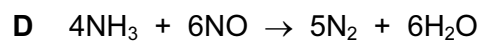
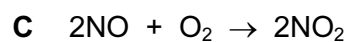
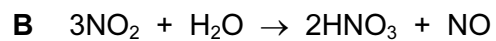
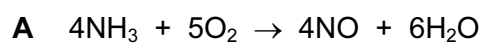
- 10 Which row correctly describes the electrodes used in the electrolysis cell for the production of aluminium?

	anode	cathode
A	carbon	carbon
B	carbon	steel
C	steel	carbon
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2 In which reaction does a single nitrogen atom have the greatest change in oxidation number?



Question 16

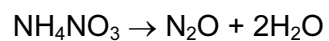
9701_w13_qp_13 • Q3

- 3 A 10 cm^3 sample of 0.30 mol dm^{-3} Tl^+NO_3^- required 20 cm^3 of 0.10 mol dm^{-3} acidified NH_4VO_3 to oxidise it to Tl^{3+} in solution. Vanadium is the only element reduced in this reaction.

What is the oxidation number of the vanadium in the reduced form?

- A** +1 **B** +2 **C** +3 **D** +4

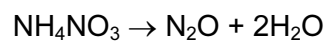
- 6 Ammonium nitrate, NH_4NO_3 , can decompose explosively when heated.



What are the changes in the oxidation numbers of the two nitrogen atoms in NH_4NO_3 when this reaction proceeds?

- A** -2, -4 **B** +2, +6 **C** +4, -6 **D** +4, -4

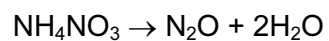
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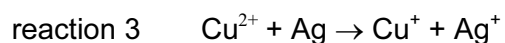
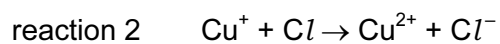
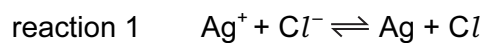


What are the changes in the oxidation numbers of the two nitrogen atoms in NH_4NO_3 when this reaction proceeds?

- A** -2, -4 **B** +2, +6 **C** +4, -6 **D** +4, -4

- 10 Photochromic glass, used for sunglasses, darkens when exposed to bright light and becomes more transparent again when the light is less bright. The depth of colour of the glass is related to the concentration of silver atoms.

The following reactions are involved.



Which statement about these reactions is correct?

- A Cu^+ and Cu^{2+} ions act as catalysts.
- B Cu^+ ions act as an oxidising agent in reaction 2.
- C Reaction 2 is the one in which light is absorbed.
- D Ag^+ ions are oxidised in reaction 1.

Answer Key

No.	Paper	Topic	Answer
1	9701_s24_qp_13	6. Electrochemistry	B
2	9701_s23_qp_13	6. Electrochemistry	D
3	9701_m22_qp_12	6. Electrochemistry	A
4	9701_w22_qp_11	6. Electrochemistry	C
5	9701_w22_qp_12	6. Electrochemistry	C
6	9701_s21_qp_13	6. Electrochemistry	D
7	9701_s19_qp_11	6. Electrochemistry	B
8	9701_w19_qp_11	6. Electrochemistry	D
9	9701_w19_qp_13	6. Electrochemistry	D
10	9701_s18_qp_11	6. Electrochemistry	B
11	9701_s17_qp_11	6. Electrochemistry	B
12	9701_s16_qp_11	6. Electrochemistry	A
13	9701_w14_qp_11	6. Electrochemistry	A
14	9701_w14_qp_12	6. Electrochemistry	A
15	9701_w13_qp_11	6. Electrochemistry	A
16	9701_w13_qp_13	6. Electrochemistry	B
17	9701_s10_qp_11	6. Electrochemistry	D
18	9701_s10_qp_12	6. Electrochemistry	D
19	9701_s10_qp_13	6. Electrochemistry	D
20	9701_w09_qp_12	6. Electrochemistry	A